

MAT11 - 3

Linear Algebra I

[Module page](#)

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About this textbook

This year, for the first time, every Bachelor course at Albert School has its own textbook, thanks to your teachers and AI.

- The textbook is a help for working on the course at home, after the class.
- The course as taught in class remains the reference, and prevails over the textbook wherever the two differ.
- This is a first version. Help us improve it by sharing your feedback with your teacher.

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Preface

By the end of this course you will be able to do a set of things you cannot do yet. You will compute in $\mathbb{R}^n$ — dot products, norms, distances, and the angle between two vectors — and bound these quantities with the Cauchy–Schwarz and triangle inequalities; you will decide when vectors are orthogonal, determine the orthogonal complement of a family, and build orthogonal or orthonormal vectors by orthogonal projection and the Gram–Schmidt process. You will solve any system of linear equations by Gaussian elimination and read off from its pivots whether the solution is unique, absent, or a whole family. You will add and multiply matrices, invert them — a 2×2 matrix by a formula, a larger one by Gauss–Jordan elimination — and tie invertibility to the kernel. You will recognise a vector space and its subspaces, describe the subspace a family generates, and prove that a set is a subspace in three standard ways. You will test a family for linear independence, extract or complete a basis, compute a dimension, and apply the incomplete-basis theorem. You will decide whether a map is linear, compute its kernel, image, and rank, and apply the rank theorem; you will represent a linear map by a matrix once bases are fixed, change bases, and recognise two similar matrices as one map seen two ways. You will decompose a space into complementary subspaces and count dimensions with Grassmann’s formula. And throughout, you will read every algebraic fact — a kernel, a rank, an invertibility — as a geometric feature of a transformation of space: what it collapses, what it preserves, what it leaves fixed.

This book assumes only what *Foundations* built — logical reasoning, sets and functions, and the rules of differentiation and integration used in a handful of examples; everything proper to linear algebra is developed here from the ground up.

1. The space $\mathbb{R}^n$

Everything in this course happens inside $\mathbb{R}^n$, or inside spaces that behave like it. So we begin here, with the one space you can **see**: for $n = 2$ and $n = 3$ its vectors are the arrows of school geometry, and the algebra we build is just those arrows written in coordinates. A row of a dataset — a customer's (age, income, spend) — is exactly such a vector; the geometry set up here is what later lets us speak of the **distance** between two data points and the **angle** between two features.

1.1. Vectors, addition, scaling

Definition 1. For an integer $n \geq 1$, the set $\mathbb{R}^n$ is the n -fold **Cartesian product** $\mathbb{R} \times \dots \times \mathbb{R}$: its elements are the ordered n -tuples

$$x = (x_1, \dots, x_n), \quad x_1, \dots, x_n \in \mathbb{R},$$

called **vectors**; the number x_i is the i -th **coordinate**. Two operations act **componentwise**: for $x, y \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$,

$$x + y = (x_1 + y_1, \dots, x_n + y_n), \quad \lambda x = (\lambda x_1, \dots, \lambda x_n).$$

The **zero vector** is $0 = (0, \dots, 0)$, and the **opposite** of x is $-x = (-1)x$.

Geometrically, in $\mathbb{R}^2$ and $\mathbb{R}^3$, a vector is an **arrow from the origin** to the point with those coordinates. Addition is the **parallelogram rule** — slide v until its tail sits at the tip of u , and $u + v$ is the arrow to where it lands; scaling by λ **stretches** the arrow by the factor λ , reversing its direction when $\lambda < 0$ (Figure 1). These pictures are the whole reason the componentwise formulas are worth caring about.

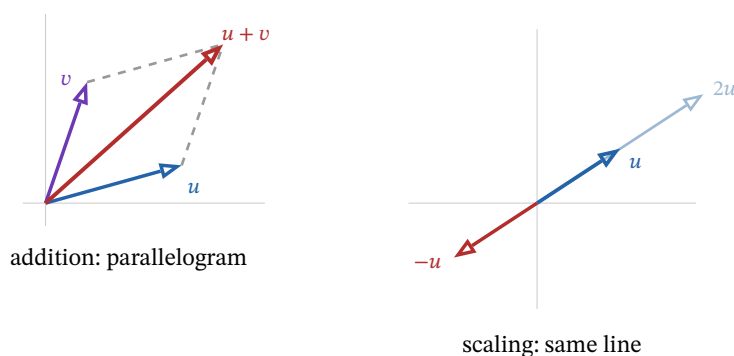


Figure 1: Vector operations in $\mathbb{R}^2$. **Left:** the sum $u + v$ (red) is the diagonal of the parallelogram built on u (blue) and v (purple) — equivalently, slide v tail-to-tip onto u . **Right:** every scalar multiple of one vector u lies on a single line through the origin; $2u$ points the same way at twice the length, $-u$ the opposite way.

Remark. These operations obey exactly the eight **vector-space rules** of Chapter 6, seen here in their first and most concrete instance. When we later add polynomials or functions, the pictures from $\mathbb{R}^2$ will still guide the algebra even though the arrows are gone.

1.2. Dot product, norm, distance

The operations so far only **combine** vectors. To **measure** — lengths, distances, angles — we need one more piece of structure: a product of two vectors that returns a **number**.

Definition 2. The **dot product** (or **Euclidean inner product**) of $x, y \in \mathbb{R}^n$ is the number

$$\langle x, y \rangle = x_1 y_1 + x_2 y_2 + \dots + x_n y_n = \sum_{i=1}^n x_i y_i.$$

The **norm** (or **length**) of x is

$$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x_1^2 + \dots + x_n^2},$$

and the **distance** between x and y is $d(x, y) = \|x - y\|$.

Elementary texts often write this product $x \cdot y$; throughout this book it is written $\langle x, y \rangle$, the standard notation for an inner product and the form the course's later results are stated in.

Proposition 1 (properties of the dot product). For all $x, y, z \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$:

- **symmetry:** $\langle x, y \rangle = \langle y, x \rangle$;
- **linearity in each slot:** $\langle x + z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$ and $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$ (and likewise in the second slot, by symmetry);
- **positivity:** $\langle x, x \rangle = \|x\|^2 \geq 0$, with $\langle x, x \rangle = 0$ if and only if $x = 0$.

Positivity is what makes $\|x\|$ a genuine length: it is never negative, and it vanishes only for the zero vector. Each property is read straight off the coordinate formula — for positivity, $\langle x, x \rangle = \sum x_i^2$ is a sum of squares, zero only when every $x_i = 0$.

Example — one pair, every quantity. Take $x = (1, 2, 2)$ and $y = (3, 0, 4)$ in $\mathbb{R}^3$. Then

$$\langle x, y \rangle = 1 \cdot 3 + 2 \cdot 0 + 2 \cdot 4 = 11, \quad \|x\| = \sqrt{1 + 4 + 4} = 3, \quad \|y\| = \sqrt{9 + 0 + 16} = 5,$$

and the distance is

$$d(x, y) = \|x - y\| = \|(-2, 2, -2)\| = \sqrt{4 + 4 + 4} = 2\sqrt{3}.$$

Pitfall. Keep the types straight: $\langle x, y \rangle$ is a **number**, λx is a **vector**, and $\|x\|$ is a **number**. In particular $\|x + y\|$ is **not** $\|x\| + \|y\|$ in general — that failed equality is exactly what the triangle inequality below measures.

The dot product also carries the **angle** between two vectors. In $\mathbb{R}^2$ and $\mathbb{R}^3$, elementary trigonometry gives

$$\langle x, y \rangle = \|x\| \|y\| \cos \theta,$$

where θ is the angle between the arrows x and y (Figure 2, left). In particular $\langle x, y \rangle = 0$ means the vectors are **orthogonal** (perpendicular): the dot product is the coordinate test for a right angle. Reading the formula the other way **defines** the angle in every $\mathbb{R}^n$ — provided the ratio $(\langle x, y \rangle)/(\|x\|\|y\|)$ always lands in $[-1, 1]$, which is precisely what the next inequality guarantees.

Example — the angle for the running pair. For the same $x = (1, 2, 2)$ and $y = (3, 0, 4)$, the three quantities above were $\langle x, y \rangle = 11$, $\|x\| = 3$ and $\|y\| = 5$. The angle θ between them therefore satisfies

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\|\|y\|} = \frac{11}{15}, \quad \text{so} \quad \theta = \arccos\left(\frac{11}{15}\right) \approx 42.8^\circ.$$

The ratio $11/15 \approx 0.733$ lies in $[-1, 1]$, as it must.

1.3. Cauchy–Schwarz and the triangle inequalities

Two inequalities govern norms and distances. The first bounds a dot product by the product of the lengths; the second says a straight path is the shortest.

Theorem 1 (Cauchy–Schwarz inequality). For all $x, y \in \mathbb{R}^n$,

$$|\langle x, y \rangle| \leq \|x\|\|y\|,$$

with equality if and only if x and y are **collinear** (one is a scalar multiple of the other).

Proof. The inequality. If $y = 0$ both sides are 0 and it holds. Otherwise consider the real function

$$\varphi(t) = \|x + ty\|^2 = \langle x + ty, x + ty \rangle = \|y\|^2 t^2 + 2\langle x, y \rangle t + \|x\|^2,$$

expanded using the linearity and symmetry of the dot product. This is a quadratic in t with positive leading coefficient $\|y\|^2 > 0$, and by positivity $\varphi(t) \geq 0$ for every t . A quadratic that is never negative has discriminant ≤ 0 :

$$(2\langle x, y \rangle)^2 - 4\|y\|^2\|x\|^2 \leq 0 \implies (\langle x, y \rangle)^2 \leq \|x\|^2\|y\|^2.$$

Taking square roots gives $|\langle x, y \rangle| \leq \|x\|\|y\|$.

The equality case, both directions. Suppose first that equality holds. If $y = 0$ the vectors are already collinear; if $y \neq 0$, equality makes the discriminant vanish, so φ has a (double) root t_0 with $\|x + t_0 y\|^2 = \varphi(t_0) = 0$, hence $x + t_0 y = 0$, i.e. $x = -t_0 y$: the vectors are collinear. Conversely, suppose x and y are collinear. Then either $y = 0$, where both sides equal 0, or $x = \lambda y$ for some $\lambda \in \mathbb{R}$, and then

$$|\langle x, y \rangle| = |\lambda\|y\|^2| = (|\lambda|\|y\|)\|y\| = \|x\|\|y\|,$$

using $\|x\| = |\lambda|\|y\|$. So equality holds **exactly** when x and y are collinear. $\square$

Theorem 2 (triangle inequalities). For all $x, y \in \mathbb{R}^n$,

$$\|x + y\| \leq \|x\| + \|y\| \quad (\text{triangle inequality}),$$

and consequently

$$\| \|x\| - \|y\| \| \leq \|x - y\| \quad (\text{reverse triangle inequality}).$$

Proof. Expanding and bounding the middle term with Cauchy–Schwarz,

$$\|x + y\|^2 = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.$$

Both sides are ≥ 0 , so taking square roots gives the triangle inequality. For the reverse form, apply it to $x = (x - y) + y$: this gives $\|x\| \leq \|x - y\| + \|y\|$, hence $\|x\| - \|y\| \leq \|x - y\|$; swapping x and y gives $\|y\| - \|x\| \leq \|x - y\|$, and the two together are $\| \|x\| - \|y\| \| \leq \|x - y\|$. $\square$

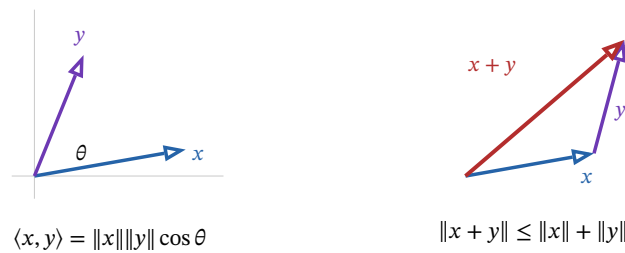


Figure 2: **Left:** the dot product encodes the angle, $\langle x, y \rangle = \|x\|\|y\| \cos \theta$; the two vectors are orthogonal exactly when $\langle x, y \rangle = 0$. **Right:** the triangle inequality says the straight arrow $x + y$ (red) is no longer than the bent path along x then y — a straight line is the shortest route between two points.

Remark. Both quantities are read straight off feature vectors in data science. The distance $d(x, y)$ measures how **far apart** two data points are; the **cosine similarity** $(\langle x, y \rangle) / (\|x\|\|y\|)$ — the cosine of the angle, pinned to $[-1, 1]$ by Cauchy–Schwarz — measures how **aligned** two directions are, and is the standard score for comparing documents or users regardless of their magnitude.

1.4. Orthogonality and orthonormal families

The equation $\langle x, y \rangle = 0$ says the two arrows meet at a right angle. Made systematic, this single equation organises whole families of vectors and gives $\mathbb{R}^n$ the rigid, grid-like structure the rest of the course exploits.

Definition 3. Two vectors $x, y \in \mathbb{R}^n$ are **orthogonal**, written $x \perp y$, when $\langle x, y \rangle = 0$. A family $(u_1, \dots, u_k)$ of vectors of $\mathbb{R}^n$ is **orthogonal** when its vectors are non-zero and pairwise orthogonal,

$$\langle u_i, u_j \rangle = 0 \quad \text{for all } i \neq j,$$

and **orthonormal** when moreover each vector has norm 1, i.e. $\|u_i\| = 1$ for every i .

The zero vector is orthogonal to every vector, itself included, so it is excluded from an orthogonal family on purpose. Orthonormality is only a rescaling: dividing each u_i of an orthogonal family by its norm $\|u_i\|$ produces an orthonormal family pointing the same ways.

Example — the standard basis is orthonormal. In $\mathbb{R}^n$ the vectors $e_1, \dots, e_n$, where e_i carries a 1 in position i and 0 elsewhere, satisfy $\langle e_i, e_j \rangle = 0$ for $i \neq j$ and $\langle e_i, e_i \rangle = 1$; so $(e_1, \dots, e_n)$ is orthonormal. In $\mathbb{R}^2$ the pair $((1, 1), (1, -1))$ is orthogonal — its dot product is $1 - 1 = 0$ — and dividing each vector by its norm $\sqrt{2}$ makes it orthonormal.

Orthogonality earns its own name because it forces a rigidity ordinary families lack: within an orthogonal family, no vector is a linear combination of the others.

Proposition 2 (an orthogonal family has no redundancy). Let $(u_1, \dots, u_k)$ be an orthogonal family in $\mathbb{R}^n$. If

$$\lambda_1 u_1 + \dots + \lambda_k u_k = 0$$

for some scalars $\lambda_1, \dots, \lambda_k \in \mathbb{R}$, then $\lambda_1 = \dots = \lambda_k = 0$.

Proof. Fix an index $j \in \llbracket 1, k \rrbracket$ and take the dot product of the relation with u_j . By the linearity of the dot product in its first slot,

$$0 = \left\langle \sum_{i=1}^k \lambda_i u_i, u_j \right\rangle = \sum_{i=1}^k \lambda_i \langle u_i, u_j \rangle = \lambda_j \langle u_j, u_j \rangle = \lambda_j \|u_j\|^2,$$

because $\langle u_i, u_j \rangle = 0$ for every $i \neq j$. Since $u_j \neq 0$, its norm $\|u_j\|$ is non-zero, so $\lambda_j = 0$. The index j was arbitrary, so every coefficient vanishes. $\square$

This is exactly the property named **linear independence** in Chapter 8: an orthogonal family is the easiest kind to certify independent. The right angle also restores the Pythagorean theorem, now in every dimension.

Proposition 3 (Pythagorean theorem). If $x, y \in \mathbb{R}^n$ are orthogonal, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$.

Proof. By the linearity and symmetry of the dot product,

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2.$$

Orthogonality gives $\langle x, y \rangle = 0$, leaving $\|x + y\|^2 = \|x\|^2 + \|y\|^2$. $\square$

1.5. The orthogonal complement of a family

Given some vectors, the vectors orthogonal to **all** of them form a set worth naming and computing.

Definition 4. The **orthogonal complement** of a family $(a_1, \dots, a_k)$ of vectors of $\mathbb{R}^n$ is the set of vectors orthogonal to every one of them,

$$\{a_1, \dots, a_k\}^\perp = \{x \in \mathbb{R}^n : \langle x, a_i \rangle = 0 \text{ for all } i \in \llbracket 1, k \rrbracket\}.$$

Each condition $\langle x, a_i \rangle = 0$ is one linear equation in the coordinates $x_1, \dots, x_n$ of x , so the orthogonal complement is the common solution set of k homogeneous equations — a **linear system**, solved by the elimination of Chapter 3.

Example — orthogonal complement of two vectors in space. Take $a = (1, 2, 2)$ and $b = (2, 1, 0)$ in $\mathbb{R}^3$. A vector $x = (x_1, x_2, x_3)$ lies in $\{a, b\}^\perp$ exactly when

$$\langle x, a \rangle = x_1 + 2x_2 + 2x_3 = 0 \quad \text{and} \quad \langle x, b \rangle = 2x_1 + x_2 = 0.$$

The second equation gives $x_2 = -2x_1$; substituting into the first, $x_1 - 4x_1 + 2x_3 = 0$, hence $x_3 = \frac{3}{2}x_1$. So

$$x = x_1 \left(1, -2, \frac{3}{2} \right) \in \text{Span}((2, -4, 3)),$$

and $\{a, b\}^\perp$ is the **line** spanned by $(2, -4, 3)$. One checks $\langle (2, -4, 3), a \rangle = 2 - 8 + 6 = 0$ and $\langle (2, -4, 3), b \rangle = 4 - 4 = 0$.

Remark. The orthogonal complement of a family is **always** a subspace of $\mathbb{R}^n$ (Chapter 7): stacking the a_i as the rows of a matrix A , the condition “ $\langle x, a_i \rangle = 0$ for all i ” reads $Ax = 0$, so $\{a_1, \dots, a_k\}^\perp$ is the kernel of A — and every kernel is a subspace. Determining it is one homogeneous system away.

1.6. Orthogonal projection and the Gram–Schmidt process

Dropping a perpendicular from a point onto a line is the oldest construction in geometry; in $\mathbb{R}^n$ it is one formula.

Definition 5. Let $a \in \mathbb{R}^n$ be non-zero. The **orthogonal projection** of $x \in \mathbb{R}^n$ onto the line $\mathbb{R}a = \{ta : t \in \mathbb{R}\}$ is the vector

$$p_a(x) = \frac{\langle x, a \rangle}{\langle a, a \rangle} a.$$

The coefficient $(\langle x, a \rangle)/(\langle a, a \rangle)$ is chosen for exactly one reason: it makes the **remainder** $x - p_a(x)$ orthogonal to a .

Proposition 4 (a projection splits off an orthogonal remainder). For non-zero $a \in \mathbb{R}^n$ and any $x \in \mathbb{R}^n$, the remainder $x - p_a(x)$ is orthogonal to a . Hence x splits as

$$x = \underbrace{p_a(x)}_{\text{along } a} + \underbrace{(x - p_a(x))}_{\perp a},$$

a multiple of a plus a vector orthogonal to a .

Proof. Take the dot product of the remainder with a , using linearity:

$$\langle x - p_a(x), a \rangle = \langle x, a \rangle - \frac{\langle x, a \rangle}{\langle a, a \rangle} (\langle a, a \rangle) = \langle x, a \rangle - \langle x, a \rangle = 0.$$

So $x - p_a(x)$ is orthogonal to a ; adding and subtracting $p_a(x)$ gives the displayed splitting. □

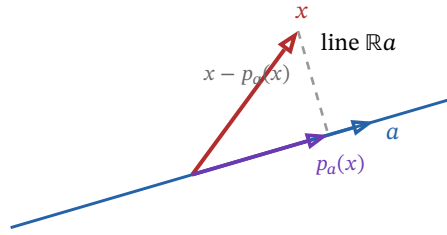


Figure 3: Orthogonal projection of x (red) onto the line $\mathbb{R}a$ (blue). The foot $p_a(x)$ (purple) is the point of the line closest to x , and the dashed remainder $x - p_a(x)$ meets the line at a right angle. This single “keep the perpendicular part” step is the atom of the Gram–Schmidt process below.

Iterating this move — split off the part along a direction, keep the perpendicular remainder — turns **any** family of vectors into an orthogonal one. This is the **Gram–Schmidt process**.

Method 1 (Gram–Schmidt process). From vectors $a_1, \dots, a_k \in \mathbb{R}^n$, define $u_1, \dots, u_k$ in turn:

$$u_1 = a_1, \quad u_j = a_j - \sum_{i=1}^{j-1} \frac{\langle a_j, u_i \rangle}{\langle u_i, u_i \rangle} u_i \quad (2 \leq j \leq k).$$

The subtracted sum is the projection of a_j onto the earlier directions, so u_j is the part of a_j perpendicular to all of $u_1, \dots, u_{j-1}$. If no a_j is a linear combination of $a_1, \dots, a_{j-1}$, then every u_j is non-zero and $(u_1, \dots, u_k)$ is an **orthogonal** family with the same linear combinations as $(a_1, \dots, a_k)$. Dividing each u_j by $\|u_j\|$ makes the family **orthonormal**.

Proof (the output family is orthogonal). We prove by induction on m that $(u_1, \dots, u_m)$ is orthogonal. For $m = 1$ there is nothing to check. Assume $(u_1, \dots, u_{m-1})$ is orthogonal and fix $l < m$. Taking the dot product of the defining formula for u_m with u_l ,

$$\langle u_m, u_l \rangle = \langle a_m, u_l \rangle - \sum_{i=1}^{m-1} \frac{\langle a_m, u_i \rangle}{\langle u_i, u_i \rangle} \langle u_i, u_l \rangle.$$

By the induction hypothesis $\langle u_i, u_l \rangle = 0$ for every $i \neq l$, so only the $i = l$ term survives, and

$$\langle u_m, u_l \rangle = \langle a_m, u_l \rangle - \frac{\langle a_m, u_l \rangle}{\langle u_l, u_l \rangle} \langle u_l, u_l \rangle = 0.$$

Thus u_m is orthogonal to every earlier u_l , so $(u_1, \dots, u_m)$ is orthogonal. By induction the whole family $(u_1, \dots, u_k)$ is orthogonal. □

That $(u_1, \dots, u_k)$ produces the same linear combinations as $(a_1, \dots, a_k)$ is read off the formula both ways: each u_j is a_j minus a combination of earlier u_i , and each a_j is u_j plus a combination of earlier u_i . Neither rewriting reaches outside the vectors already built.

Example — Gram–Schmidt on two vectors. Orthogonalise $a_1 = (1, 1, 0)$ and $a_2 = (1, 0, 1)$ in $\mathbb{R}^3$. First $u_1 = a_1 = (1, 1, 0)$. Then strip from a_2 its projection onto u_1 , using $\langle a_2, u_1 \rangle = 1$ and $\langle u_1, u_1 \rangle = 2$:

$$u_2 = a_2 - \frac{\langle a_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right).$$

A check confirms the right angle: $\langle u_1, u_2 \rangle = \frac{1}{2} - \frac{1}{2} + 0 = 0$. Normalising with $\|u_1\| = \sqrt{2}$ and $\|u_2\| = \sqrt{3/2}$,

$$\left(\frac{1, 1, 0}{\sqrt{2}}, \frac{1, -1, 2}{\sqrt{6}} \right)$$

is an orthonormal family with the same linear combinations as (a_1, a_2) .

Pitfall. In the Gram–Schmidt sum, always project onto the **already-orthogonalised** vectors u_i , never onto the raw inputs a_i . Subtracting projections onto the a_i leaves the remainders correlated and breaks orthogonality. The process is sequential for exactly this reason: u_j depends on every u_i built before it.

2. Matrix arithmetic

We turn to the object you can hold in your hand and compute with directly. A **matrix** is a rectangular table of numbers; everything abstract later in the course is, ultimately, a statement about matrices in disguise.

2.1. Definition and notation

A **matrix** of size $m \times n$ is a rectangular array of real numbers with m rows and n columns,

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix},$$

written compactly $A = (a_{ij})$, where a_{ij} is the entry in **row** i , **column** j . The set of all such matrices is written $\mathcal{M}_{m,n}(\mathbb{R})$, and $\mathcal{M}_n(\mathbb{R})$ when $m = n$ (the **square** matrices). A matrix with one column ($n = 1$) is a **column vector**; with one row ($m = 1$), a **row vector**. To save vertical space we write a column vector as a **transposed row**, $(x_1, \dots, x_n)^\top$ — throughout this book the superscript $\top$ is only this space-saving shorthand, meaning “the column with entries $x_1, \dots, x_n$ ”.

Pitfall. The index order is **row first, then column** — always. The entry a_{23} lives in row 2, column 3, never the reverse. A matrix of size $m \times n$ has m rows and n columns, in that order. Getting this backwards is the single most common source of confusion in matrix multiplication below.

2.2. The operations

Before the general rules, compute one concrete instance of each.

Example — the operations on small matrices. Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & -1 \\ 2 & 5 \end{pmatrix}$. Then

$$A + B = \begin{pmatrix} 5 & 1 \\ 2 & 8 \end{pmatrix}, \quad 3A = \begin{pmatrix} 3 & 6 \\ 0 & 9 \end{pmatrix}.$$

For the product, each entry of AB is a row of A “dotted” against a column of B :

$$AB = \begin{pmatrix} 1 \cdot 4 + 2 \cdot 2 & 1 \cdot (-1) + 2 \cdot 5 \\ 0 \cdot 4 + 3 \cdot 2 & 0 \cdot (-1) + 3 \cdot 5 \end{pmatrix} = \begin{pmatrix} 8 & 9 \\ 6 & 15 \end{pmatrix}.$$

Definition 6. Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices.

- **Addition** (defined only when A and B have the **same size**): $(A + B)_{ij} = a_{ij} + b_{ij}$.
- **Scalar multiplication**: for $\lambda \in \mathbb{R}$, $(\lambda A)_{ij} = \lambda a_{ij}$.
- **Matrix multiplication** (defined only when A is $m \times n$ and B is $n \times p$, so the inner sizes match): AB is the $m \times p$ matrix

$$(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

The product entry $(AB)_{ij}$ pairs the i -th **row** of A with the j -th **column** of B , multiplying corresponding entries and summing. For this pairing to make sense the row of A and the column of B must have the same length: **the number of columns of A must equal the number of rows of B .**

Pitfall. Matrix multiplication is the operation students rush. Two traps: (i) the product AB is only defined when the inner dimensions agree — $(m \times n)(n \times p)$ — and its size is the **outer** dimensions $m \times p$; (ii) you sum **across** a row of A and **down** a column of B , not entry by matching entry. When in doubt, write the dimensions under the matrices and cancel the inner pair.

2.3. Algebraic properties — and the one that fails

Proposition 5. For matrices of compatible sizes and scalars $\lambda, \mu \in \mathbb{R}$:

- addition is **commutative** and **associative**, with the zero matrix as identity element;
- scalar multiplication distributes: $\lambda(A + B) = \lambda A + \lambda B$ and $(\lambda + \mu)A = \lambda A + \mu A$;
- multiplication is **associative**, $(AB)C = A(BC)$, and **distributes** over addition, $A(B + C) = AB + AC$.

Every one of these matches your arithmetic instincts — **except** one crucial absence from the list. Matrix multiplication is **not commutative**.

Proposition 6. In general $AB \neq BA$.

Proof. By counterexample: one failing instance refutes a universal claim. Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

so $AB \neq BA$. One pair suffices to refute the universal claim. □

Pitfall. Non-commutativity is not a curiosity — it is a fact you must **actively remember** in every manipulation. You may not “swap factors” as with numbers. In particular $(A + B)^2 = A^2 + AB + BA + B^2$, which is **not** $A^2 + 2AB + B^2$ unless $AB = BA$.

2.4. The identity matrix

The $n \times n$ **identity matrix** I_n has 1 on the main diagonal and 0 everywhere else,

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is the neutral element of multiplication: $AI_n = A$ and $I_nA = A$ whenever the product is defined. It plays the role the number 1 plays for ordinary multiplication, and it will reappear as the matrix of the “do-nothing” linear map.

3. Linear systems

The oldest use of matrices is to **solve simultaneous linear equations**. This chapter turns the schoolbook task of solving such a system into a systematic algorithm whose by-products — pivots, free variables — become the backbone of the whole course.

3.1. Matrix form

A system of m linear equations in n unknowns $x_1, \dots, x_n$,

$$\begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n = b_m \end{cases},$$

collapses into a single **matrix equation**

$$AX = B,$$

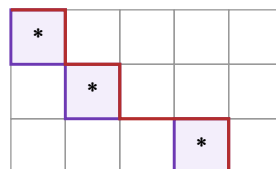
where $A = (a_{ij}) \in \mathcal{M}_{m,n}(\mathbb{R})$ is the **coefficient matrix**, $X = (x_1, \dots, x_n)^\top$ the column of unknowns, and $B = (b_1, \dots, b_m)^\top$ the column of right-hand sides. The product AX , read with the rule of Chapter 2, reproduces the left-hand sides exactly — which is the whole reason matrix multiplication is defined the way it is.

3.2. Gaussian elimination

The method is the one you already use by hand — eliminate a variable, repeat — made systematic. We work on the **augmented matrix** $(A \mid B)$, the coefficient matrix with the right-hand side appended as an extra column, and we are allowed three **elementary row operations**, each of which transforms the system into an **equivalent** one (same solution set):

1. **swap** two rows;
2. **scale** a row by a non-zero constant;
3. **add** a multiple of one row to another.

The goal is **row echelon form**: in every non-zero row the first non-zero entry — the **pivot** — sits strictly to the right of the pivot in the row above, and any all-zero rows are pushed to the bottom. The pivots then trace a descending “staircase”.



pivots descend left-to-right

Figure 4: Row echelon form: each pivot (purple, marked *) lies strictly to the right of the one above it, so the pivots descend in a staircase (red). Columns **with** a pivot correspond to pivot variables; columns **without** one — here the 3rd and 5th — correspond to free variables.

Example — elimination to echelon form. Solve

$$\begin{cases} x + 2y + z = 4 \\ 2x + 5y - z = 3 \\ x + 3y - 2z = -1 \end{cases}.$$

The augmented matrix, then $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 4 & 4 \\ 2 & 5 & -1 & 3 & 3 \\ 1 & 3 & -2 & -1 & -1 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 1 & 4 & 4 \\ 0 & 1 & -3 & -5 & -5 \\ 0 & 1 & -3 & -5 & -5 \end{array} \right).$$

Now $R_3 \leftarrow R_3 - R_2$ wipes out the third row entirely:

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 1 & 4 & 4 \\ 0 & 1 & -3 & -5 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right).$$

Pivots sit in columns 1 and 2; column 3 has none, so z is **free**. Back-substituting, $y = -5 + 3z$ and $x = 4 - 2y - z = 14 - 7z$. The solution set is a line, treated next.

3.3. Pivot analysis and solution classification

After reduction, the pivot pattern decides everything. A column holding a pivot is a **pivot variable**; a column without one is a **free variable**. Then:

- if any row reads $0 = c$ with $c \neq 0$ (a pivot in the augmented column), the system is **inconsistent** — **no solution**;
- otherwise, if **every** variable is a pivot variable, the solution is **unique**;
- otherwise there are free variables, and the system has **infinitely many solutions**, one for each assignment of the free variables.

Pitfall. Count pivots **in the coefficient part**, but always check the augmented column separately. A pivot in the right-hand column alone signals the contradiction $0 = c$ and means **no solution** — even if the coefficient part looked fine. Many “I found infinitely many solutions” errors are really an overlooked inconsistency.

The worked system above landed on the infinitely-many case; the other two outcomes are just as short to reduce.

Example — a unique solution, and no solution. Unique solution. For

$$\begin{cases} x + y = 3 \\ 2x + y = 4 \end{cases} \quad \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 1 & 4 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -1 & -2 \end{array} \right)$$

(using $R_2 \leftarrow R_2 - 2R_1$). Both columns of the coefficient part carry a pivot, so there is no free variable. Back-substituting, the last row gives $-y = -2$, hence $y = 2$, and then $x = 3 - y = 1$. The system has the **unique** solution $(x, y) = (1, 2)$.

No solution. For

$$\begin{cases} x + y = 1 \\ 2x + 2y = 3 \end{cases}, \quad \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 3 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 1 \end{array} \right)$$

(again $R_2 \leftarrow R_2 - 2R_1$). The last row reads $0 = 1$: a pivot stands alone in the augmented column, so the system is **inconsistent** and has no solution — exactly the case the pitfall above warns about, now produced by a reduction.

3.4. Parametric form

When free variables exist, name them as **parameters** and express each pivot variable in terms of them. The solution set then reads as a **particular solution plus a span of directions**:

$$X = X_0 + t_1 V_1 + \dots + t_k V_k, \quad t_1, \dots, t_k \in \mathbb{R},$$

where X_0 is one solution and $V_1, \dots, V_k$ are the directions traced by the free variables.

Example — the previous system in parametric form. With $z = t$ free, $x = 14 - 7t$ and $y = -5 + 3t$, so

$$X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 14 \\ -5 \\ 0 \end{pmatrix} + t \begin{pmatrix} -7 \\ 3 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

This is a line in $\mathbb{R}^3$ through the point $(14, -5, 0)$ in direction $(-7, 3, 1)$. The vector $(-7, 3, 1)$ is exactly the free-variable direction; notice it is what you get by setting $t = 1$ and dropping the constant — this reappears as a **basis of the null space** in Section 7.3.

4. A matrix acting on a vector

We now reinterpret the symbol AX . So far it was a bookkeeping device for a system; now read it as a **machine** that eats a vector and emits a vector. This shift — from “array of numbers” to “transformation” — is the conceptual hinge of the course.

A matrix $A \in \mathcal{M}_{m,n}(\mathbb{R})$ defines a map

$$\begin{aligned} f_A : \mathbb{R}^n &\longrightarrow \mathbb{R}^m \\ X &\longmapsto AX \end{aligned}$$

sending each input column X (with n entries) to the output AX (with m entries). The defining feature of this map is that it **respects linear combinations**:

$$A(X + Y) = AX + AY, \quad A(\lambda X) = \lambda(AX).$$

A map with this property is called **linear** — the central notion of Chapter 10. For now, two subspaces attached to A deserve names, because the entire theory to come is built to compute them.

Definition 7. For $A \in \mathcal{M}_{m,n}(\mathbb{R})$:

- the **null space** (or **kernel**) is the set of inputs sent to zero,

$$\ker A = \{X \in \mathbb{R}^n : AX = 0\};$$

- the **column space** (or **image**) is the set of all outputs,

$$\operatorname{Im} A = \{AX : X \in \mathbb{R}^n\},$$

which equals the set of all linear combinations of the **columns** of A .

The two are exactly the two sides of solving $AX = B$. The null space is the solution set of the **homogeneous** system $AX = 0$ — the free-variable directions of Chapter 3. The column space is the set of right-hand sides B for which $AX = B$ is **solvable at all**: B is reachable precisely when it is a combination of the columns. Reading “ $AX = B$ has a solution” as “ B lies in the column space” is a habit worth forming now.

Example — reading off the column space. For $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, both columns point along $(1, 2)$, so every output AX is a multiple of $(1, 2)$: the column space is the **line** spanned by $(1, 2)$, not all of $\mathbb{R}^2$. Hence $AX = B$ is solvable only for B on that line — e.g. solvable for $B = (3, 6)$, impossible for $B = (1, 0)$.

5. Invertible matrices

For an ordinary number $a \neq 0$ there is a reciprocal a^{-1} that undoes multiplication by a . We now ask the same of a **square** matrix: is there a matrix that undoes A ? When there is, solving $AX = B$ becomes a single multiplication, and — the payoff of this chapter — the same Gaussian elimination of Chapter 3 both finds the inverse and decides whether one exists.

5.1. The inverse of a square matrix

Definition 8. A square matrix $A \in \mathcal{M}_n(\mathbb{R})$ is **invertible** if there is a matrix $B \in \mathcal{M}_n(\mathbb{R})$ with

$$AB = BA = I_n.$$

Such a B is unique; it is the **inverse** of A , written A^{-1} . A matrix with no inverse is called **singular**.

Proof (uniqueness of the inverse). If B and C both satisfy the definition, then, using associativity of the matrix product,

$$\begin{aligned} B &= BI_n = B(AC) = (BA)C \\ &= I_n C = C. \end{aligned}$$

So the inverse, when it exists, is unique. □

Only square matrices can be invertible: AB and BA must both be defined and equal to an identity, which forces A and B to be square of the same size. Two rules are used constantly.

Proposition 7. If $A, B \in \mathcal{M}_n(\mathbb{R})$ are invertible, then so are A^{-1} and AB , with

$$(A^{-1})^{-1} = A, \quad (AB)^{-1} = B^{-1}A^{-1}.$$

The reversal in $(AB)^{-1} = B^{-1}A^{-1}$ has a simple logic: to undo “first B , then A ”, you undo A first, then B . Check it by multiplying: $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I_n$.

5.2. The 2×2 formula

For the smallest square matrices there is a closed formula.

Proposition 8 (inverse of a 2×2 matrix). Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If the number $ad - bc$ is **non-zero**, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If $ad - bc = 0$, then A is **not** invertible.

Proof. Multiply out:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix} = (ad - bc)I_2.$$

When $ad - bc \neq 0$, dividing by it gives $A \cdot \left(\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\right) = I_2$, and the same computation in the other order gives I_2 too, so the displayed matrix is A^{-1} . When $ad - bc = 0$, the two columns of A are collinear, so $\ker A \neq \{0\}$ (some non-zero X has $AX = 0$); by the theorem in Section 5.4 below, such an A cannot be invertible. $\square$

Example — a 2×2 inverse. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $ad - bc = 2 \cdot 1 - 1 \cdot 1 = 1$, so

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Check: $AA^{-1} = \begin{pmatrix} 2-1 & -2+2 \\ 1-1 & -1+2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. ✓

Remark. The number $ad - bc$ is the exact gatekeeper of invertibility in the 2×2 case, and it will return in *Linear Algebra II* as one instance of a single quantity attached to any square matrix. Here we neither name it nor develop it further — the 2×2 formula is all we need.

5.3. Inverting by Gauss–Jordan elimination

Beyond 2×2 there is no formula worth memorising; instead we **compute** the inverse by the elimination we already know. The recipe:

Method 2 (Gauss–Jordan inversion). To invert $A \in \mathcal{M}_n(\mathbb{R})$, form the **augmented** $n \times 2n$ matrix $(A \mid I_n)$ and apply elementary row operations until the left block is I_n . The right block is then A^{-1} :

$$(A \mid I_n) \sim \dots \sim (I_n \mid A^{-1}).$$

If at any stage the left block acquires a **zero row**, it can never become I_n : then A is **not invertible**, and the process stops.

Why it works: every row operation is achieved by **left-multiplying** by some invertible matrix. The sequence of operations that turns A into I_n therefore left-multiplies A by A^{-1} — so applying that **same** sequence to I_n turns it into $A^{-1}I_n = A^{-1}$. Carrying I_n alongside simply records the product.

Example — a 3×3 inverse by Gauss–Jordan. Invert $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$. Reduce $(A \mid I_3)$; first clear column 1 with $R_3 \leftarrow R_3 - 5R_1$, then column 2 below the pivot with $R_3 \leftarrow R_3 + 4R_2$:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 5 & 6 & 0 & 0 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array} \right).$$

The left block is now upper-triangular with pivots 1, 1, 1; clear **upward** with $R_2 \leftarrow R_2 - 4R_3$, $R_1 \leftarrow R_1 - 3R_3$, then $R_1 \leftarrow R_1 - 2R_2$:

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 16 & -12 & -3 \\ 0 & 1 & 0 & 20 & -15 & -4 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -24 & 18 & 5 \\ 0 & 1 & 0 & 20 & -15 & -4 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array} \right).$$

The left block is I_3 , so

$$A^{-1} = \begin{pmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{pmatrix}.$$

Check the top-left entry of AA^{-1} — row 1 of A against column 1 of A^{-1} : $(1, 2, 3) \cdot (-24, 20, -5) = -24 + 40 - 15 = 1$, and the other entries fall out the same way to give I_3 . ✓

5.4. Kernel and invertibility

Whether the Gauss–Jordan process **succeeds** is decided entirely by the kernel of A — this is the promised link between solving $AX = 0$ and inverting A .

Theorem 3 (invertibility criteria). For a square matrix $A \in \mathcal{M}_n(\mathbb{R})$, the following are equivalent:

1. A is invertible;
2. $\ker A = \{0\}$ (the homogeneous system $AX = 0$ has only the solution $X = 0$);
3. A has a pivot in **every** one of its n columns after row reduction;
4. for every B , the system $AX = B$ has a **unique** solution.

Proof. $(1 \implies 2)$ If A is invertible and $AX = 0$, then $X = A^{-1}AX = A^{-1}0 = 0$, so $\ker A = \{0\}$. $(2 \iff 3)$ A column without a pivot is a free variable, giving a non-zero solution of $AX = 0$; so $\ker A = \{0\}$ exactly when every column has a pivot. $(3 \implies 4)$ With n pivots in n columns and n rows, back-substitution solves $AX = B$ for every B , uniquely (no free variables). $(4 \implies 1)$ First, condition 4 with $B = 0$ says $AX = 0$ has a **unique** solution; since $X = 0$ is a solution, it is the only one, so $\ker A = \{0\}$. Now solve $AX = e_j$ for each standard basis vector e_j — condition 4 provides a solution C_j — and let C be the matrix with columns $C_1, \dots, C_n$, so that $AC = I_n$. It remains to check $CA = I_n$: from $AC = I_n$,

$$A(CA - I_n) = (AC)A - A = A - A = 0,$$

so every column of $CA - I_n$ lies in $\ker A = \{0\}$ and hence vanishes, giving $CA = I_n$. Thus A is invertible with $A^{-1} = C$ — which is exactly what Gauss–Jordan computes. $\square$

Remark. Read geometrically (the thread of this whole course), A is invertible when the transformation $X \mapsto AX$ **collapses no direction**: its kernel is trivial, so distinct inputs stay distinct and every target is hit exactly once — the map is a bijection of $\mathbb{R}^n$, undone by A^{-1} . A singular matrix crushes some non-zero vector to 0, losing the information needed to reverse it.

Pitfall. Invertibility is a property of **square** matrices only. A 2×3 or 3×2 matrix is never invertible — there is no single matrix undoing it on both sides. And for a square A , $ad - bc =$

0 (in the 2×2 case) or a zero row appearing in Gauss–Jordan (in general) is the signal of singularity, not a computational mistake to fix.

6. Vector spaces

Here is the leap promised in the preface. We have been adding and scaling several different kinds of objects — numerical tuples, polynomials, matrices, functions. We now extract the **rules** they all obey and take those rules as a definition. Anything satisfying them is a **vector space**, and any theorem we prove from the rules alone holds for all of them simultaneously.

6.1. From a familiar example to the axioms

In $\mathbb{R}^n$ you add componentwise and scale componentwise, and these operations satisfy a short list of identities — addition commutes and associates, there is a zero, every vector has a negative, and scalar multiplication distributes. We promote that list to a definition.

Definition 9. A **vector space over $\mathbb{R}$** is a set E equipped with an **addition** $E \times E \longrightarrow E$ and a **scalar multiplication** $\mathbb{R} \times E \longrightarrow E$ such that, for all $u, v, w \in E$ and all $\lambda, \mu \in \mathbb{R}$:

1. addition is **associative**: $(u + v) + w = u + (v + w)$;
2. addition is **commutative**: $u + v = v + u$;
3. there is a **zero vector** 0 with $u + 0 = u$;
4. every u has an **opposite** $-u$ with $u + (-u) = 0$;
5. $\lambda(u + v) = \lambda u + \lambda v$;
6. $(\lambda + \mu)u = \lambda u + \mu u$;
7. $(\lambda\mu)u = \lambda(\mu u)$;
8. $1 \cdot u = u$.

The elements of E are called **vectors**.

Remark. The axioms are a **recognition checklist**, not something to memorize for its own sake. The intended reading is: “ $\mathbb{R}^n$, with the operations you have used since school, satisfies these — so let us agree to call anything else that satisfies them a vector space too, and reuse all our results there.”

6.2. The standard examples

Example — the spaces this course lives in. Each of the following is a vector space over $\mathbb{R}$; in each, addition and scaling are the obvious componentwise/pointwise operations, and the zero vector is named.

- $\mathbb{R}^n$ — numerical vectors; zero is $(0, \dots, 0)$.
- $\mathbb{R}[X]$ — all polynomials, and $\mathbb{R}_n[X]$ — polynomials of degree $\leq n$; zero is the zero polynomial.
- $\mathcal{F}(\mathbb{R}, \mathbb{R})$ — all functions $\mathbb{R} \longrightarrow \mathbb{R}$, with $(f + g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda f(x)$; zero is the constant function 0 .
- $\mathcal{M}_{m,n}(\mathbb{R})$ — matrices of a fixed size; zero is the zero matrix.

Pitfall. Watch the degree example. The polynomials of degree **exactly** n do **not** form a vector space: their sum can have lower degree (e.g. $(X^2 + X) + (-X^2) = X$), and they contain no zero polynomial. It is $\mathbb{R}_n[X]$, degree **at most** n , that is a space. “At most” is closed under addition; “exactly” is not.

In the examples above the operations were the familiar ones, so the axioms held “obviously”. To **verify** the axioms is a skill in its own right — and it earns its keep exactly when the operations are unfamiliar, because then nothing can be taken for granted. Here is such a case, checked from scratch.

Example — a from-scratch verification: same rules, new operations. Let $E = \mathbb{R}_{>0}$ be the **positive** reals, with an addition and a scaling that are **not** the usual ones:

$$u \boxplus v = uv \quad (\text{ordinary product}), \quad \lambda \boxdot u = u^\lambda.$$

Nothing is inherited from $\mathbb{R}$, so we check the eight axioms directly. Addition $\boxplus$ is associative and commutative because multiplication of reals is. The **zero vector** is the number 1, since $u \boxplus 1 = u \cdot 1 = u$; the **opposite** of u is $1/u$, since $u \boxplus (1/u) = 1$, the zero vector. For the four scaling axioms,

$$\lambda \boxdot (u \boxplus v) = (uv)^\lambda = u^\lambda v^\lambda = (\lambda \boxdot u) \boxplus (\lambda \boxdot v),$$

$$(\lambda + \mu) \boxdot u = u^{\lambda+\mu} = u^\lambda u^\mu = (\lambda \boxdot u) \boxplus (\mu \boxdot u),$$

$$(\lambda\mu) \boxdot u = u^{\lambda\mu} = (u^\mu)^\lambda = \lambda \boxdot (\mu \boxdot u), \quad 1 \boxdot u = u^1 = u.$$

All eight hold, so $(\mathbb{R}_{>0}, \boxplus, \boxdot)$ is a vector space — one whose zero vector is 1 and where “adding” means multiplying. When the operations are unfamiliar there is no shortcut: each axiom is settled by unwinding the definitions, exactly as here.

7. Subspaces

In practice one rarely checks all eight axioms from scratch. Instead, most spaces of interest sit **inside** a space already known to be a vector space — the solutions of $AX = 0$ inside $\mathbb{R}^n$, the polynomials of degree ≤ 2 inside $\mathbb{R}[X]$. For such **subsets**, a three-line test replaces the whole axiom list.

7.1. The three-criterion test

Definition 10. A subset F of a vector space E is a **subspace** of E if it is itself a vector space under the operations of E .

Theorem 4 (Three-criterion test). A subset $F \subseteq E$ is a subspace if and only if

1. $0 \in F$ (it contains the zero vector);
2. F is **closed under addition**: $u, v \in F \implies u + v \in F$;
3. F is **closed under scalar multiplication**: $\lambda \in \mathbb{R}$ and $u \in F \implies \lambda u \in F$.

The test works because the eight axioms are **inherited** automatically from E : associativity, commutativity and the rest already hold for all of E , hence for its elements in F . The only things that could go wrong are that F misses the zero vector or “leaks” outside itself under the operations — and that is exactly what the three criteria forbid. Criteria 2 and 3 are often merged: F is a subspace iff $0 \in F$ and $\lambda u + v \in F$ for all $\lambda \in \mathbb{R}, u, v \in F$.

The geometric content of criterion 1 is worth a picture: a subspace must pass through the origin.

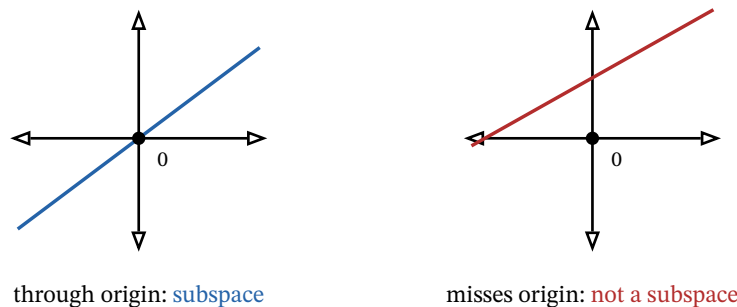


Figure 5: Every subspace contains the zero vector, so in $\mathbb{R}^2$ the only subspaces are $\{0\}$, lines **through the origin**, and the whole plane. A line that misses the origin (right) fails criterion 1 immediately — a quick first test before checking closure.

Example — applying the test. Is $F = \{(x, y, z) \in \mathbb{R}^3 : x + y - 2z = 0\}$ a subspace of $\mathbb{R}^3$? (1) $(0, 0, 0)$ satisfies the equation, so $0 \in F$. (2) If (x, y, z) and (x', y', z') both satisfy it, adding the two equations shows their sum does too. (3) Scaling an equation by λ preserves it. All three hold, so F is a subspace. Contrast $G = \{(x, y, z) : x + y - 2z = 1\}$: here $0 \notin G$, so G fails at criterion 1 and is **not** a subspace.

Pitfall. A solution set of a **homogeneous** system $AX = 0$ is always a subspace; a solution set of an **inhomogeneous** system $AX = B$ with $B \neq 0$ never is (it misses the origin). This is exactly the distinction between the null space and a parametric solution $X_0 + (\dots)$.

7.2. Span: the subspace generated by a set

Given some vectors, the smallest subspace containing them is obtained by closing them under the two operations — that is, by forming all their linear combinations.

Definition 11. The **span** (or **generated subspace**) of vectors $v_1, \dots, v_k \in E$ is

$$\text{Span}(v_1, \dots, v_k) = \{\lambda_1 v_1 + \dots + \lambda_k v_k : \lambda_i \in \mathbb{R}\}.$$

It is a subspace of E , and it is the **smallest** subspace containing all of $v_1, \dots, v_k$.

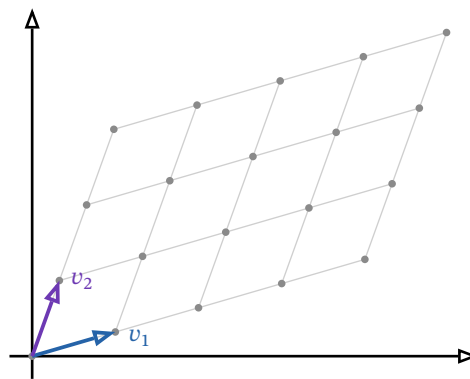


Figure 6: The span of two non-parallel vectors v_1, v_2 in $\mathbb{R}^2$. The grey dots are the integer combinations $iv_1 + jv_2$; they sit on a regular lattice whose two families of lines run along v_1 and v_2 . Allowing **real** coefficients fills in every point between the dots, so the span is the entire plane — two non-parallel vectors already generate $\mathbb{R}^2$.

7.3. Column space and null space by row reduction

For a matrix A the column space and null space (Chapter 4) are subspaces, and **one row reduction computes both**:

- the **null space** is read from the parametric solution of $AX = 0$ — the free-variable direction vectors span it;
- the **column space** is spanned by the columns of A sitting in **pivot positions** (the pivot columns of the original A).

This is the first sign that row reduction is the universal computational engine: the abstract questions “what is the null space?” and “what is the column space?” are answered by the elimination you already do by hand.

Example — both spaces from a single reduction. Take $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{pmatrix}$. One elimination, $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$, then $R_3 \leftarrow R_3 - R_2$, gives the echelon form

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

with pivots in columns 1 and 3; column 2 carries no pivot, so y is free.

Null space. Solving $AX = 0$ on the echelon form: the second row gives $z = 0$, the first gives $x + 2y + z = 0$, so $x = -2y$. Hence $(x, y, z) = y(-2, 1, 0)$ and

$$\ker A = \text{Span}((-2, 1, 0)), \quad \dim \ker A = 1.$$

Column space. The pivots sit in columns 1 and 3, so the **original** first and third columns form a basis:

$$\text{Im } A = \text{Span}((1, 2, 3), (1, 3, 4)), \quad \dim \text{Im } A = 2.$$

One reduction delivered both, and the count checks out: the 3 columns split as nullity 1 + rank 2.

7.4. Three standard ways to prove a subspace

We now have every tool the course uses to establish that a subset is a subspace. Which one you reach for depends only on **how the subset is described**.

To prove that a subset $F \subseteq E$ is a subspace, use whichever fits:

1. **The criteria.** Verify directly that $0 \in F$, and that F is closed under addition and under scalar multiplication (the three-criterion test above). Use this when F is given by a **condition** on its elements.
2. **As a span.** Exhibit an explicit family with $F = \text{Span}(v_1, \dots, v_k)$; a span is automatically a subspace. Use this when you can **list generators** of F .
3. **As a kernel.** Recognise F as the solution set of a **homogeneous** system, i.e. $F = \ker A$ for a matrix A (and, from Chapter 10 on, $F = \ker f$ for a linear map f); every such set is a subspace. Use this when F is cut out by linear equations **equal to zero**.

These are three views of one fact, and they often convert into one another: the plane $F = \{(x, y, z) : x + y - 2z = 0\}$ is at once a **condition** (method 1), a **span** $\text{Span}((1, 0, 1/2), (0, 1, 1/2))$ (method 2), and the **kernel** of the 1×3 matrix $\begin{pmatrix} 1 & 1 & -2 \end{pmatrix}$ (method 3). Reading a set all three ways is a core fluency of the course.

7.5. Sum, intersection, and direct sum

Definition 12. For subspaces F, G of E :

- the **intersection** $F \cap G = \{u : u \in F \text{ and } u \in G\}$ is a subspace;
- the **sum** $F + G = \{u + v : u \in F, v \in G\}$ is a subspace — the smallest one containing both F and G .

The sum is **direct**, written $F \oplus G$, when every vector of $F + G$ writes in **exactly one** way as $u_F + u_G$ with $u_F \in F$ and $u_G \in G$. When moreover the direct sum fills the whole space, $F \oplus$

$G = E$, the subspaces F and G are **complementary** (or **supplementary**) in E : every $u \in E$ then decomposes uniquely as $u = u_F + u_G$.

Unique decomposition is the whole point of a direct sum, and a single condition on the intersection detects it.

Theorem 5 (Characterization of a direct sum). For subspaces F, G of E :

1. the sum $F + G$ is direct if and only if $F \cap G = \{0\}$;
2. consequently F and G are complementary — $E = F \oplus G$ — if and only if

$$F \cap G = \{0\} \quad \text{and} \quad E = F + G.$$

Proof. (1) Suppose the sum is direct, and let $w \in F \cap G$. Then w has the two decompositions $w = w + 0$ (with $w \in F, 0 \in G$) and $w = 0 + w$ (with $0 \in F, w \in G$); uniqueness forces $w = 0$, so $F \cap G = \{0\}$. Conversely suppose $F \cap G = \{0\}$ and some vector has two decompositions $u_F + u_G = u'_F + u'_G$. Rearranging, $u_F - u'_F = u'_G - u_G$; the left side lies in F and the right side in G , so this common vector lies in $F \cap G = \{0\}$. Hence $u_F = u'_F$ and $u_G = u'_G$: the decomposition is unique and the sum is direct.

(2) By definition F and G are complementary exactly when the sum is direct **and** equals E . Substituting “ $F \cap G = \{0\}$ ” for “direct” via part (1) gives the stated pair of conditions. $\square$

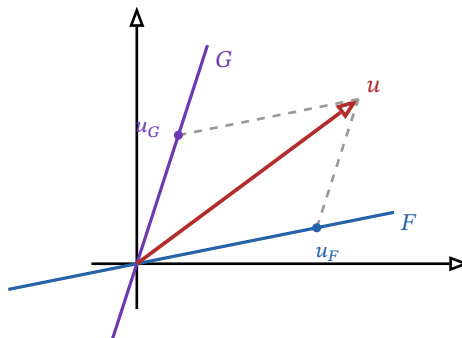


Figure 7: Two distinct lines through the origin are complementary subspaces of $\mathbb{R}^2$: $F \oplus G = \mathbb{R}^2$. Each vector u (red) has a **unique** decomposition $u = u_F + u_G$ found by projecting along the two lines (the dashed parallelogram). Uniqueness fails the instant the lines coincide, i.e. the instant $F \cap G \neq \{0\}$.

8. Linear independence, bases, dimension

We can now answer the question lurking behind op(“Span”): **how many vectors do you really need** to generate a space, and is there redundancy among a given set? The answers — independence, basis, dimension — are the quantitative heart of linear algebra.

8.1. Linear independence

A set of vectors is **redundant** if one of them is a combination of the others; removing it changes nothing about the span. **Independence** is the absence of any such redundancy, captured by a single clean equation.

Definition 13. Vectors $v_1, \dots, v_k$ are **linearly independent** if the only scalars satisfying

$$\lambda_1 v_1 + \dots + \lambda_k v_k = 0$$

are $\lambda_1 = \dots = \lambda_k = 0$ (the **trivial** combination). If some non-trivial combination gives 0, they are **linearly dependent**.

Example — testing independence is solving a homogeneous system. Are $v_1 = (1, 1, 0)$, $v_2 = (0, 1, 1)$, $v_3 = (1, 0, -1)$ independent? Set $\lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 = 0$; componentwise this is the homogeneous system with these vectors as **columns**. Row-reducing finds $\lambda_1 = -\lambda_3$, $\lambda_2 = \lambda_3$, with λ_3 free — a non-trivial solution (take $\lambda_3 = -1$, giving $(\lambda_1, \lambda_2, \lambda_3) = (1, -1, -1)$). So they are **dependent**; indeed $v_1 - v_2 - v_3 = 0$. Independence questions are always homogeneous systems in disguise.

Pitfall. Independence is a property of the **whole family**, not of pairs. Three vectors can be pairwise non-parallel yet still dependent — as above, where no two are multiples of each other but the three together satisfy $v_1 - v_2 - v_3 = 0$. Always test the full combination, never just “are any two parallel?”.

8.2. Bases and dimension

Definition 14. Let E be a vector space.

- A family is **generating** (or **spanning**) if its span is all of E .
- A **basis** of E is a family that is **both** linearly independent **and** generating.

A basis is the “just right” family: generating means **enough** vectors to reach everything, independent means **no more than enough**. Equivalently — and this is the property to remember — a basis is a family in which **every** vector of E writes **uniquely** as a linear combination of the basis vectors.

Why should the count of “just right” be the same for every basis? One lemma settles it, and it is worth seeing proved — it is the engine behind every dimension count in the course.

Lemma 1 (Exchange lemma). If $u_1, \dots, u_p$ are linearly independent in E and $w_1, \dots, w_q$ generate E , then $p \leq q$: an independent family is never larger than a generating one.

Proof. We feed the independent vectors into the generating family one at a time, each time discarding a w and keeping the family generating. Since $w_1, \dots, w_q$ generate E , write $u_1 = \sum_i a_i w_i$; the coefficients are not all zero (otherwise $u_1 = 0$, impossible in an independent family), say $a_1 \neq 0$. Solving for w_1 shows w_1 is a combination of $u_1, w_2, \dots, w_q$, so this new family still generates E . Suppose after k steps the family $u_1, \dots, u_k, w_{k+1}, \dots, w_q$ still generates. Then u_{k+1} is a combination of it; the coefficients of the remaining w 's cannot all vanish — else u_{k+1} would be a combination of $u_1, \dots, u_k$, contradicting independence — so we discard one such w and insert u_{k+1} . If $p > q$ we would run out of w 's after q steps, leaving $u_1, \dots, u_q$ generating E ; then u_{q+1} would be a combination of them, contradicting independence. Hence $p \leq q$. $\square$

Theorem 6 (Dimension is well-defined). In a finitely generated vector space, all bases have the **same** number of vectors. This common number is the **dimension** of E , written $\dim E$.

Proof. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases, of sizes p and q . Since $\mathcal{B}$ is independent and $\mathcal{B}'$ generates, the lemma gives $p \leq q$. Swapping their roles — $\mathcal{B}'$ independent, $\mathcal{B}$ generating — gives $q \leq p$. Hence $p = q$. $\square$

Proposition 9 (Incomplete-basis theorem). In a finite-dimensional space E : every linearly independent family can be **extended** to a basis, and every generating family **contains** a basis.

Proof. To extend an independent family, keep adjoining any vector lying **outside** its current span; each addition preserves independence and enlarges the span, so the process must halt — and it can only halt when the span is all of E , i.e. at a basis. To thin a generating family, discard any vector that is already a combination of the others; the family keeps generating, and when no such vector remains it is independent — a basis. $\square$

Example — standard dimensions. The **standard basis** of $\mathbb{R}^n$ is $e_1, \dots, e_n$ with e_i having a 1 in position i and 0 elsewhere; thus $\dim \mathbb{R}^n = n$. Likewise $\dim \mathbb{R}_n[X] = n + 1$ (basis $1, X, \dots, X^n$) and $\dim \mathcal{M}_{m,n}(\mathbb{R}) = mn$.

8.3. Reading dimension off the pivots

The connection to Chapter 3 is the practical payoff of the whole course. Form the matrix whose columns are your vectors and row-reduce:

Row reduction reads independence, basis, and dimension.

- the vectors are **independent** iff **every** column is a pivot column (no free variables);
- the **pivot columns of the original matrix** form a basis of their span;
- the **rank** of the matrix — the number of pivots — equals the **dimension** of that span.

So “extract a basis from a generating family” is just “keep the pivot columns”, and “compute the dimension of a span” is just “count the pivots”. The abstract notions of basis and dimension are computed by elimination.

The **same** pivots run the reverse task — the **completion** direction of the incomplete-basis theorem, turning an independent family into a basis by adding vectors to it.

Example — completing an independent family to a basis. Extend the independent family $(1, 1, 0)$ to a basis of $\mathbb{R}^3$. Place it **first**, adjoin the whole standard basis e_1, e_2, e_3 behind it, and reduce the matrix whose columns are $(1, 1, 0), e_1, e_2, e_3$:

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Pivots fall in columns 1, 2 and 4; column 3 has none. Keeping the **original** pivot columns and dropping the redundant e_2 leaves

$$(1, 1, 0), \quad e_1 = (1, 0, 0), \quad e_3 = (0, 0, 1),$$

three independent vectors — a basis of $\mathbb{R}^3$ that **contains the vector we started from**. Putting the starting family first guarantees its columns become pivots, so they survive; the adjoined e_i only fill the gaps. This is completion made concrete: adjoin a spanning set, keep the pivot columns.

Proposition 10 (Counting in a space of known dimension). In a space E with $\dim E = n$:

- any linearly independent family has **at most** n vectors;
- any generating family has **at least** n vectors;
- any family of **exactly** n vectors that is independent **or** generating is automatically a **basis**.

The last point saves enormous work: once you know $\dim E = n$, a candidate basis of n vectors needs only **one** of the two properties checked, not both.

9. Grassmann's formula

Sums and intersections of subspaces interact through a single, memorable counting law — the linear-algebra analogue of inclusion–exclusion for sets. It carries the name of Hermann Grassmann, who founded the theory of dimension in the 1840s.

Theorem 7 (Grassmann's formula). For finite-dimensional subspaces F, G of a vector space,

$$\dim(F + G) = \dim F + \dim G - \dim(F \cap G).$$

Because it relates the **four** dimensions $\dim(F + G)$, $\dim F$, $\dim G$, $\dim(F \cap G)$, it is also known as the **four-dimensions formula**. The intuition mirrors counting a union of two sets: adding $\dim F$ and $\dim G$ counts the overlap $F \cap G$ **twice**, so we subtract it once. It is the tool for finding the dimension of an intersection that is awkward to describe directly: rearrange to $\dim(F \cap G) = \dim F + \dim G - \dim(F + G)$.

Proof. Start from a basis $u_1, \dots, u_r$ of $F \cap G$ (so $\dim(F \cap G) = r$). Because $F \cap G$ sits inside both F and G , the incomplete-basis theorem lets us extend it — to a basis $u_1, \dots, u_r, f_1, \dots, f_s$ of F (so $\dim F = r + s$) and to a basis $u_1, \dots, u_r, g_1, \dots, g_t$ of G (so $\dim G = r + t$). We claim the combined family $u_1, \dots, u_r, f_1, \dots, f_s, g_1, \dots, g_t$ is a basis of $F + G$. It generates $F + G$, since together the two bases reach all of F and all of G . For independence, suppose

$$\sum_i a_i u_i + \sum_j b_j f_j + \sum_k c_k g_k = 0.$$

Then $\sum_k c_k g_k = -(\sum_i a_i u_i + \sum_j b_j f_j)$ lies in F (right-hand side) and in G (left-hand side), hence in $F \cap G$. Writing this vector in the basis $u_1, \dots, u_r$ and comparing with $\sum_k c_k g_k$ inside the basis $u_1, \dots, u_r, g_1, \dots, g_t$ of G forces every $c_k = 0$. What remains, $\sum_i a_i u_i + \sum_j b_j f_j = 0$, is a relation in the basis of F , so all a_i and b_j vanish too. The family is independent, thus a basis, and

$$\begin{aligned} \dim(F + G) &= r + s + t \\ &= (r + s) + (r + t) - r \\ &= \dim F + \dim G - \dim(F \cap G). \end{aligned}$$

□

Example — two planes in space. Let F, G be two distinct planes through the origin in $\mathbb{R}^3$, so $\dim F = \dim G = 2$. Their sum is all of $\mathbb{R}^3$ (two distinct planes together reach everything), so $\dim(F + G) = 3$. Then

$$\dim(F \cap G) = 2 + 2 - 3 = 1 :$$

two distinct planes through the origin meet in a **line**, exactly as geometry insists.

Proposition 11 (Direct sum, by the numbers). $F \oplus G = E$ holds if and only if both $F \cap G = \{0\}$ **and** $\dim F + \dim G = \dim E$.

This is the practical test for complementary subspaces: check that the dimensions add up to $\dim E$ and that the intersection is trivial. Indeed, when $F \cap G = \{0\}$ Grassmann's formula gives $\dim(F + G) = \dim F + \dim G = \dim E$, so $F + G = E$; together with $F \cap G = \{0\}$ this is exactly complementarity by the characterization of a direct sum in Chapter 7.

Pitfall. Both conditions are needed. Two distinct lines in $\mathbb{R}^3$ satisfy $F \cap G = \{0\}$ but only $\dim F + \dim G = 2 \neq 3$, so they do **not** span $\mathbb{R}^3$. Conversely the dimensions can add up while the intersection is non-trivial. Neither condition alone certifies a direct-sum decomposition.

10. Linear maps

We promoted the rules of $\mathbb{R}^n$ to the abstract notion of a vector space. Now we do the same for the **maps** between them. The map $X \mapsto AX$ of Chapter 4 respected linear combinations; we take that property as the definition of the maps worth studying.

10.1. Definition, kernel, image

Definition 15. A map $f : E \longrightarrow F$ between vector spaces is **linear** if for all $u, v \in E$ and $\lambda \in \mathbb{R}$,

$$f(u + v) = f(u) + f(v), \quad f(\lambda u) = \lambda f(u).$$

Equivalently, f commutes with all linear combinations: $f(\lambda u + \mu v) = \lambda f(u) + \mu f(v)$.

Example — linear and non-linear. The map $f : \mathbb{R}^2 \longrightarrow \mathbb{R}^2, (x, y) \mapsto (2x - y, x + 3y)$, is linear — each output coordinate is a linear combination of the inputs, so $f(X) = AX$ with $A = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$. The map $g : \mathbb{R}^2 \longrightarrow \mathbb{R}^2, (x, y) \mapsto (x + 1, y)$, is **not** linear: it sends 0 to $(1, 0) \neq 0$, and a linear map must fix the zero vector (take $\lambda = 0$). Differentiation $p \mapsto p'$ and integration $f \mapsto \int_0^1 f$ are linear too — but on those spaces linearity must be **checked** from the axioms, not read off a matrix; we do so next.

Pitfall. A linear map **must** send 0 to 0: setting $\lambda = 0$ gives $f(0) = 0$. So any map with a non-zero constant term — an “affine” map like $x \mapsto ax + b$ with $b \neq 0$ — is **not** linear, despite the school usage of “linear” for straight lines. Checking $f(0) = 0$ is the fastest disqualifier.

When a map is given by an explicit formula in coordinates, reading off the matrix A settles linearity at a glance. But when the map acts on **polynomials** or **functions**, no matrix is in sight, and linearity must be verified **directly from the two axioms** — exactly the from-scratch discipline used for the vector-space axioms in Chapter 6.

Example — verifying linearity from the axioms. Take differentiation $D : \mathbb{R}_n[X] \longrightarrow \mathbb{R}_n[X], p \mapsto p'$. To prove it linear we check the two axioms using only the differentiation rules from *Foundations*. For all polynomials p, q and all $\lambda \in \mathbb{R}$,

$$D(p + q) = (p + q)' = p' + q' = D(p) + D(q),$$

since the derivative of a sum is the sum of the derivatives, and

$$D(\lambda p) = (\lambda p)' = \lambda p' = \lambda D(p),$$

since a constant factor pulls out of a derivative. Both axioms hold, so D is linear. Integration $T : \mathcal{C}([0, 1], \mathbb{R}) \longrightarrow \mathbb{R}, f \mapsto \int_0^1 f$, is linear by the very same move: $T(f + g) = \int_0^1 (f + g) = \int_0^1 f + \int_0^1 g = T(f) + T(g)$ and $T(\lambda f) = \lambda T(f)$ are the linearity of the integral. On these spaces the axioms are the **only** tool — and they are enough.

Definition 16. For a linear map $f : E \longrightarrow F$:

- the **kernel** is $\ker f = \{u \in E : f(u) = 0\} \subseteq E$;
- the **image** is $\operatorname{Im} f = \{f(u) : u \in E\} \subseteq F$;
- the **rank** is $\operatorname{rank} f = \dim(\operatorname{Im} f)$.

Both of these sets are subspaces — a fact used on almost every page that follows, so we prove it once, straight from the three-criterion test of Chapter 7.

Proposition 12 (the kernel and image are subspaces). For a linear map $f : E \longrightarrow F$, the kernel $\ker f$ is a subspace of E and the image $\operatorname{Im} f$ is a subspace of F .

Proof. Kernel. A linear map fixes zero, $f(0) = 0$, so $0 \in \ker f$. If $u, v \in \ker f$ then $f(u + v) = f(u) + f(v) = 0 + 0 = 0$, so $u + v \in \ker f$; and for $\lambda \in \mathbb{R}$, $f(\lambda u) = \lambda f(u) = \lambda \cdot 0 = 0$, so $\lambda u \in \ker f$. The three criteria hold, so $\ker f$ is a subspace of E .

Image. Since $0 = f(0)$, we have $0 \in \operatorname{Im} f$. If $y = f(u)$ and $y' = f(v)$ lie in $\operatorname{Im} f$, then $y + y' = f(u) + f(v) = f(u + v)$ is again an output, so $y + y' \in \operatorname{Im} f$; and $\lambda y = \lambda f(u) = f(\lambda u) \in \operatorname{Im} f$. The three criteria hold, so $\operatorname{Im} f$ is a subspace of F . $\square$

These generalize the null space and column space of a matrix verbatim: for $f(X) = AX$, $\ker f = \ker A$ and $\operatorname{Im} f = \operatorname{Im} A$. Everything you learned to compute by row reduction transfers directly.

10.2. The rank theorem

The single most important theorem about a linear map says its domain splits cleanly into “what gets crushed to zero” and “what survives as the image”.

Theorem 8 (Rank theorem (rank-nullity)). For a linear map $f : E \longrightarrow F$ with E finite-dimensional,

$$\dim E = \dim(\ker f) + \operatorname{rank} f.$$

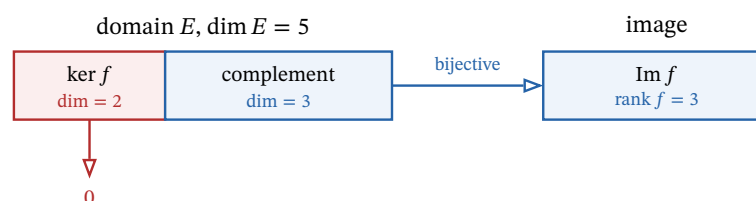


Figure 8: The rank theorem as a budget. The domain E (here $\dim E = 5$) splits into the part crushed to zero — $\ker f$, of dimension 2 — and a complement that maps **bijectively** onto the image, of dimension $\operatorname{rank} f = 3$. The dimensions must add up: $5 = 2 + 3$. Enlarging the kernel necessarily shrinks the rank, and vice versa.

The theorem is a conservation law: the $\dim E$ “degrees of freedom” in the domain are spent either on directions that vanish (the kernel) or on directions that survive into the image (the rank), with none lost. For a matrix $A \in \mathcal{M}_{m,n}(\mathbb{R})$ it reads $n = \dim(\ker A) + \operatorname{rank} A$, which is just the pivot/free-variable split of Chapter 3: rank = number of pivots, $\operatorname{nullity}$ = number of free variables, and together they account for all n columns.

Proof. Let $k = \dim(\ker f)$ and choose a basis $e_1, \dots, e_k$ of $\ker f$. By the incomplete-basis theorem, extend it to a basis $e_1, \dots, e_k, e_{k+1}, \dots, e_n$ of E , where $n = \dim E$. We show the $n - k$ vectors $f(e_{k+1}), \dots, f(e_n)$ form a basis of $\operatorname{Im} f$; then $\operatorname{rank} f = n - k$, which is the claim.

They generate $\operatorname{Im} f$: any output is $f(u)$ for some $u = \sum_{i=1}^n x_i e_i$, and since $f(e_i) = 0$ for $i \leq k$,

$$f(u) = \sum_{i=k+1}^n x_i f(e_i).$$

They are independent: if $\sum_{i=k+1}^n c_i f(e_i) = 0$, then by linearity $f(\sum_{i=k+1}^n c_i e_i) = 0$, so $\sum_{i=k+1}^n c_i e_i \in \ker f$, a combination of $e_1, \dots, e_k$. This is a linear relation among the basis $e_1, \dots, e_n$, so all its coefficients vanish — in particular every $c_i = 0$. Hence $f(e_{k+1}), \dots, f(e_n)$ is a basis of $\operatorname{Im} f$ and $\operatorname{rank} f = n - k = \dim E - \dim(\ker f)$. $\square$

10.3. Injectivity, surjectivity, bijectivity

Linearity turns the function-theoretic notions of *Foundations* into **dimension counts** — this is where the abstraction pays its way.

Proposition 13. Let $f : E \longrightarrow F$ be linear with E, F finite-dimensional.

- f is **injective** $\iff \ker f = \{0\} \iff \operatorname{rank} f = \dim E$.
- f is **surjective** $\iff \operatorname{Im} f = F \iff \operatorname{rank} f = \dim F$.
- f is **bijective** $\iff$ both hold; this forces $\dim E = \dim F$.

The injectivity criterion is the workhorse: instead of checking “ $f(u) = f(v) \implies u = v$ ” for all pairs, you check the **single** equation $f(u) = 0 \implies u = 0$. The reason is linearity — $f(u) = f(v)$ rearranges to $f(u - v) = 0$, so distinct inputs collide exactly when the kernel is non-trivial.

Proof (injective $\iff$ trivial kernel). If f is injective then $f(u) = 0 = f(0)$ forces $u = 0$, so $\ker f = \{0\}$. Conversely, suppose $\ker f = \{0\}$ and $f(u) = f(v)$. By linearity $f(u - v) = f(u) - f(v) = 0$, so $u - v \in \ker f = \{0\}$, giving $u = v$. Hence f is injective. $\square$

Corollary 1. For a linear map $f : E \longrightarrow F$ between spaces of **equal finite dimension** ($\dim E = \dim F$), injectivity, surjectivity and bijectivity are **equivalent**.

Proof. By the rank theorem $\dim(\ker f) = \dim E - \operatorname{rank} f$. Injectivity means $\dim(\ker f) = 0$, i.e. $\operatorname{rank} f = \dim E = \dim F$, which is exactly surjectivity. So each implies the other, and either one implies bijectivity. $\square$

Pitfall. That equivalence holds **only when the dimensions are equal** (in particular for a square matrix). In general a tall thin map can be injective without being surjective, and a short fat one surjective without being injective. Always confirm $\dim E = \dim F$ before invoking “injective $\implies$ bijective”.

10.4. Composition and the space $\mathcal{L}(E, F)$

Linear maps combine in two ways, and both stay inside the linear world.

Proposition 14 (composition of linear maps). If $f : E \longrightarrow F$ and $g : F \longrightarrow G$ are linear, then $g \circ f : E \longrightarrow G$ is linear. Moreover, if A is the matrix of f and B the matrix of g in chosen bases, then the matrix of $g \circ f$ is the **product** BA .

Proof. For $u, v \in E$ and $\lambda \in \mathbb{R}$, using linearity of f then of g ,

$$(g \circ f)(u + v) = g(f(u) + f(v)) = g(f(u)) + g(f(v)) = (g \circ f)(u) + (g \circ f)(v),$$

and $(g \circ f)(\lambda u) = g(\lambda f(u)) = \lambda(g \circ f)(u)$. So $g \circ f$ is linear. The matrix claim is exactly why the product was defined as it was: applying BA to a coordinate column applies A , then B . $\square$

This is the deep reason matrix multiplication looks the way it does: **the product of matrices is the composition of the maps they represent**. The non-commutativity $AB \neq BA$ of Chapter 2 is then no surprise — “do f , then g ” and “do g , then f ” are genuinely different operations.

The second way to combine maps adds them pointwise.

Definition 17. The set of all linear maps from E to F is written $\mathcal{L}(E, F)$. Equipped with the pointwise operations

$$(f + g)(u) = f(u) + g(u), \quad (\lambda f)(u) = \lambda f(u),$$

it is **itself a vector space**: a sum or scalar multiple of linear maps is again linear, and the eight axioms are inherited from F . Its zero vector is the map sending every u to 0.

Remark. So the objects of this course fold back on themselves: the **maps** between two vector spaces form a vector space in their own right. Once bases are fixed, Chapter 11 shows $\mathcal{L}(E, F)$ is nothing but $\mathcal{M}_{m,n}(\mathbb{R})$ in disguise — adding maps becomes adding matrices, composing them becomes the matrix product.

10.5. Projections, and what a map leaves fixed

Three geometric readings run through this whole course: what a transformation **collapses**, what it **preserves**, and what it **leaves fixed**. The first two now have names — the **kernel** is what collapses to 0, and the **image** (measured by the **rank**) is what is preserved. The third has an equally simple home. The **fixed vectors** of a map $f : E \longrightarrow E$ are the u with $f(u) = u$; since $f(u) = u$ rewrites as $(f - \text{id}_E)(u) = 0$, they form a subspace, the kernel of $f - \text{id}_E$. The identity leaves **everything** fixed; the sharpest intermediate case is a **projection**, which displays all three readings on one map.

Definition 18. Let $E = F \oplus G$ be a direct-sum decomposition, so every $u \in E$ writes uniquely as $u = u_F + u_G$ with $u_F \in F$ and $u_G \in G$ (Chapter 7). The **projection onto F along G** is the map

$$\begin{aligned} p : E &\longrightarrow E \\ u &\longmapsto u_F \end{aligned}$$

Proposition 15 (anatomy of a projection). The projection p onto F along G is **linear**, and

$$\operatorname{Im} p = F, \quad \ker p = G, \quad p \circ p = p.$$

Moreover p **fixes F pointwise**: $p(u) = u$ if and only if $u \in F$. So the subspace p leaves fixed is exactly its image.

Proof. Linearity is the uniqueness of the decomposition: if $u = u_F + u_G$ and $v = v_F + v_G$, then $u + v = (u_F + v_F) + (u_G + v_G)$ is **the** decomposition of $u + v$, so $p(u + v) = u_F + v_F = p(u) + p(v)$, and likewise $p(\lambda u) = \lambda u_F = \lambda p(u)$. Every output $p(u) = u_F$ lies in F , and every $w \in F$ decomposes as $w + 0$, so $p(w) = w$; hence $\operatorname{Im} p = F$ and p fixes each vector of F . Next $p(u) = 0$ means $u_F = 0$, i.e. $u = u_G \in G$, so $\ker p = G$. Finally $p(u) = u_F \in F$, which p fixes, so $p(p(u)) = p(u)$: projecting again changes nothing. And if $p(u) = u$ then $u = p(u) \in \operatorname{Im} p = F$, so the fixed set is exactly F . $\square$

Example — a projection in the plane. Take $\mathbb{R}^2 = F \oplus G$ with $F = \operatorname{Span}((1, 0))$ (the x -axis) and $G = \operatorname{Span}((1, 1))$ (Figure 9). Writing $(x, y) = s(1, 0) + t(1, 1)$ gives $t = y$ and $s = x - y$, so the projection onto F along G is

$$p(x, y) = (x - y, 0), \quad \text{with matrix } \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}.$$

It **collapses** G — $p(1, 1) = (0, 0)$ — has **image** the line F , and **leaves F fixed** — $p(1, 0) = (1, 0)$. And $p \circ p = p$: the matrix satisfies $\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$.

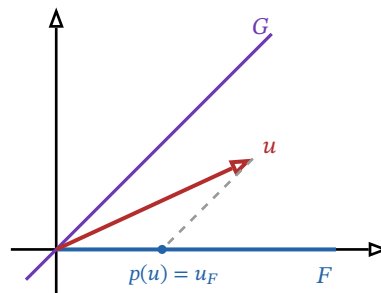


Figure 9: The projection onto F (the x -axis) along G (the line $y = x$). The vector u (red) slides down a line **parallel to G** until it lands on F . The map **collapses G** to 0 (its kernel), has **image F** (what it preserves), and **leaves every vector of F fixed** — the course's three geometric readings on a single map.

11. Matrix representation and change of basis

We close the circle. We began with matrices, abstracted to linear maps, and now show that **every** linear map between finite-dimensional spaces is, once bases are chosen, **exactly a matrix**. The abstract and the concrete are two views of one thing — and a single map, read in different bases, gives a family of **similar** matrices.

11.1. Coordinate vectors

Definition 19. Let $\mathcal{B} = (e_1, \dots, e_n)$ be a basis of E . Each $u \in E$ writes uniquely as $u = x_1 e_1 + \dots + x_n e_n$; the column

$$[u]_{\mathcal{B}} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

is the **coordinate vector** of u in the basis $\mathcal{B}$.

The uniqueness in the definition is precisely the basis property; it is what makes coordinates well-defined. Choosing a basis is choosing a **dictionary** that translates every abstract vector — a polynomial, a function — into a concrete column of numbers.

Example — coordinates depend on the basis. In $\mathbb{R}_2[X]$ with basis $\mathcal{B} = (1, X, X^2)$, the polynomial $p = 3 - X + 2X^2$ has coordinate vector $[p]_{\mathcal{B}} = (3, -1, 2)^{\top}$. In a **different** basis the same p gets a **different** column — coordinates are a property of the (vector, basis) pair, never of the vector alone.

11.2. The matrix of a linear map

Definition 20. Let $f : E \longrightarrow F$ be linear, $\mathcal{B} = (e_1, \dots, e_n)$ a basis of the source E and $\mathcal{C}$ a basis of the target F . The **matrix of f in the bases $\mathcal{B}$ and $\mathcal{C}$** , written $\text{Mat}_{\mathcal{C}, \mathcal{B}}(f)$ — the target basis first, the source basis second — is the matrix whose **j -th column** is the coordinate vector $[f(e_j)]_{\mathcal{C}}$ of the image of the j -th source basis vector.

The rule “columns are the images of the basis vectors” is worth committing to memory; it is **the** recipe for building the matrix. Its whole purpose is the following compatibility: computing with the abstract map and computing with its matrix give the same answer.

Proposition 16. With $A = \text{Mat}_{\mathcal{C}, \mathcal{B}}(f)$: if $[u]_{\mathcal{B}} = X$ then $[f(u)]_{\mathcal{C}} = AX$. Applying f to a vector becomes multiplying its coordinate column by A .

Example — matrix of differentiation. Let $f = \frac{d}{dX}$ on $\mathbb{R}_2[X]$, basis $\mathcal{B} = (1, X, X^2)$ for both source and target. The images of the basis vectors are $f(1) = 0$, $f(X) = 1$, $f(X^2) = 2X$, with coordinate columns $(0, 0, 0)^\top$, $(1, 0, 0)^\top$, $(0, 2, 0)^\top$. Stacking them as columns,

$$\text{Mat}_{\mathcal{B}}(f) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Check: $p = 3 - X + 2X^2$ has $[p] = (3, -1, 2)^\top$, and $A(3, -1, 2)^\top = (-1, 4, 0)^\top$, the coordinates of $-1 + 4X = p'$. ✓

11.3. Change of basis

A vector does not change when we re-describe it in a new basis, but its coordinate column does. The bookkeeping is handled by a single matrix.

Definition 21. Let $\mathcal{B}$ (old) and $\mathcal{B}'$ (new) be two bases of E . The **change-of-basis matrix** from $\mathcal{B}$ to $\mathcal{B}'$, written $P_{\mathcal{B} \rightarrow \mathcal{B}'}$, has, as its columns, the coordinate vectors of the **new** basis vectors $\mathcal{B}'$ expressed in the **old** basis $\mathcal{B}$. When the two bases are clear we write P for short.

Proposition 17. If $X = [u]_{\mathcal{B}}$ and $X' = [u]_{\mathcal{B}'}$ are the coordinates of the same vector u in the old and new bases, then

$$X = PX'.$$

Equivalently $X' = P^{-1}X$, so P is invertible.

Pitfall. Read the direction carefully. Although P is named “from $\mathcal{B}$ to $\mathcal{B}'$ ”, the identity $X = PX'$ means P turns **new** coordinates X' into **old** coordinates X — it transports coordinates $\mathcal{B}' \rightarrow \mathcal{B}$, the opposite way to its name. The name records where the **columns** come from (the new basis vectors, written in the old basis); the **action** on a coordinate column runs the other way. To go from old to new instead, use P^{-1} .

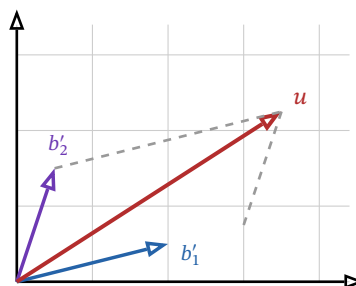


Figure 10: One vector u (red), two readings. Against the standard grid its coordinates are read off the horizontal/vertical axes; against the skewed basis (b'_1, b'_2) (blue, purple) the **same** u has coordinates from the dashed parallelogram — here $u = 1.5b'_1 + 1b'_2$. The change-of-basis matrix P converts one column of numbers into the other; the arrow itself never moves.

Example — building P and converting coordinates. In $\mathbb{R}^2$ take the old basis $\mathcal{B} = (e_1, e_2)$ (the standard basis) and the new basis $\mathcal{B}' = (b'_1, b'_2)$ with $b'_1 = (2, 1)$ and $b'_2 = (1, 1)$. Each new basis vector is already written in the old basis, so its coordinates are its own entries, and those columns **are** the change-of-basis matrix:

$$P_{\mathcal{B} \rightarrow \mathcal{B}'} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad P_{\mathcal{B}' \rightarrow \mathcal{B}}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

(the 2×2 formula, with $ad - bc = 2 \cdot 1 - 1 \cdot 1 = 1$). Now take a vector u whose coordinates in the **new** basis are $X' = (1, 2)^\top$. Its coordinates in the **old** basis are

$$X = PX' = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix},$$

matching the direct reading $u = 1 \cdot b'_1 + 2 \cdot b'_2 = (4, 3)$ in the standard basis. Conversely, a vector with old coordinates $X = (4, 3)^\top$ has new coordinates

$$X' = P^{-1}X = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

recovering X' : multiplying by P and by P^{-1} are the two directions of the same conversion.

11.4. Similar matrices

Fix a **single** linear map $f : E \rightarrow E$ of a space to itself (an **endomorphism**). Its matrix depends on the basis chosen; how do the two matrices — in $\mathcal{B}$ and in $\mathcal{B}'$ — relate? The change-of-basis matrix answers it in one line.

Proposition 18 (change of basis for an endomorphism). Let $f : E \rightarrow E$ be linear, with matrices $A = \text{Mat}_{\mathcal{B}}(f)$ and $A' = \text{Mat}_{\mathcal{B}'}(f)$ in two bases, and let P be the change-of-basis matrix from $\mathcal{B}$ to $\mathcal{B}'$. Then

$$A' = P^{-1}AP.$$

Proof. For every u the compatibility $[f(u)] = (\text{Mat})[u]$ holds in each basis, and $X = PX'$ relates the two coordinate columns. So $[f(u)]_{\mathcal{B}} = A[u]_{\mathcal{B}}$ becomes $P[f(u)]_{\mathcal{B}'} = AP[u]_{\mathcal{B}'}$, whence $[f(u)]_{\mathcal{B}'} = P^{-1}AP[u]_{\mathcal{B}'}$, for all u . That identifies the matrix of f in $\mathcal{B}'$ as $A' = P^{-1}AP$. $\square$

Definition 22. Two matrices $A, A' \in \mathcal{M}_n(\mathbb{R})$ are **similar** if $A' = P^{-1}AP$ for some invertible $P \in \mathcal{M}_n(\mathbb{R})$.

Similarity is precisely “**the same map, seen in two bases**”. Similar matrices share everything belonging to the map rather than to the accidental choice of basis — in particular the same rank and the same kernel dimension. Recognising a pair as similar means recognising two coordinate descriptions of one geometric transformation.

Example — one map, two matrices. Take $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $(x, y) \mapsto (2x + y, x - y)$; in the standard basis $\mathcal{B}$ its matrix is $A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$. Move to $\mathcal{B}' = ((1, 1), (1, -1))$. The change-of-basis matrix and its inverse are

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$A' = P^{-1}AP = \begin{pmatrix} \frac{3}{2} & \frac{3}{2} \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix},$$

which is exactly the matrix obtained by expressing $f(1, 1) = (3, 0)$ and $f(1, -1) = (1, 2)$ in the basis $\mathcal{B}'$. So A and A' are similar: one map, two bases.

Which basis makes A' **simplest** — ideally diagonal — is the central question of *Linear Algebra II*; the machinery of similar matrices built here is what that diagonalization theory runs on.